

Evaluation Report
Generated 14 June 2026 08:04

Executive Summary

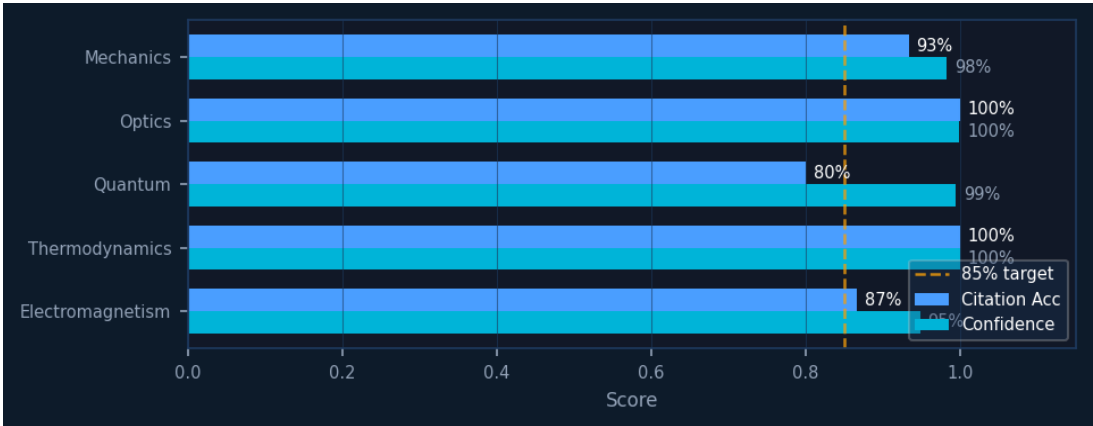
Metric	Score	Target	Status
Citation Accuracy	88.9%	≥ 85%	PASS
Hallucination Rate	0.0%	< 10%	PASS
OOS Refusal Rate	100.0%	≥ 90%	PASS
Mean Confidence	88.1%	—	—
Questions (total)	20	—	—
Physics Qs	18	—	—
OOS Qs	2	—	—

ALL TARGETS MET — PRODUCTION READY

Key Metrics

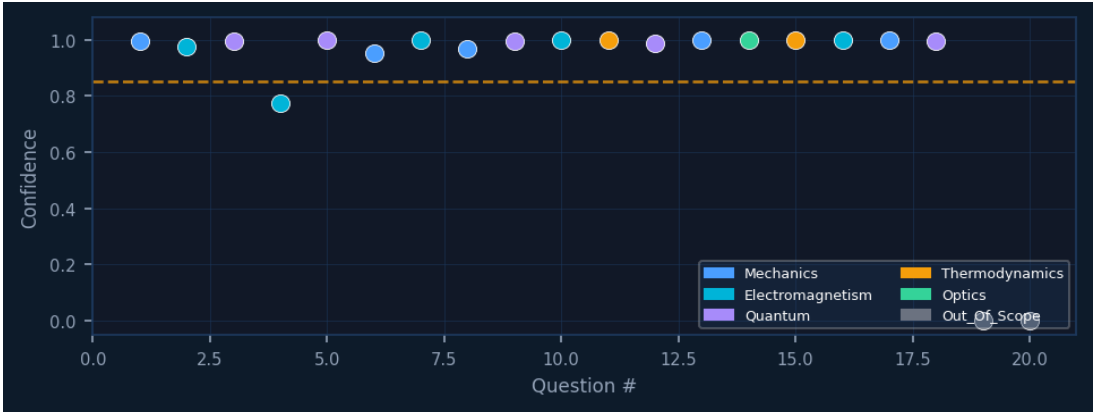


Per-Category Breakdown



Category	n	Citation Acc	Confidence	Status
Electromagnetism	5	86.7%	94.8%	PASS
Thermodynamics	2	100.0%	99.9%	PASS
Quantum	5	80.0%	99.4%	FAIL
Optics	1	100.0%	99.9%	PASS
Mechanics	5	93.3%	98.2%	PASS
Out Of Scope	2	100.0%	0.0%	OOS

Per-Question Confidence



Detailed Question Results

Each row shows the benchmark question, model answer summary, citation accuracy, confidence score, and pass/fail status.

#	Category	Question (truncated)	Cit. Acc	Conf	Status
1	Mechanics	State Newton's second law of motion and write it as an ...	100%	100%	PASS
2	Electromagnetism	What is the speed of light in a vacuum and what is its ...	100%	97%	PASS
3	Quantum	Explain the photoelectric effect and what Einstein conc...	67%	100%	WARN
4	Electromagnetism	Write Maxwell's four equations in integral form and bri...	33%	77%	WARN
5	Quantum	What is the Heisenberg uncertainty principle?	100%	100%	PASS
6	Mechanics	Derive the equation for simple harmonic motion starting...	100%	95%	PASS
7	Electromagnetism	What is Coulomb's law and what does it describe?	100%	100%	PASS
8	Mechanics	State the law of conservation of energy and give an exa...	67%	97%	WARN
9	Quantum	What is the Bohr model of the hydrogen atom and what at...	100%	100%	PASS
10	Electromagnetism	What is Faraday's law of electromagnetic induction?	100%	100%	PASS
11	Thermodynamics	Explain the first law of thermodynamics.	100%	100%	PASS
12	Quantum	What is the de Broglie wavelength and what does it impl...	100%	99%	PASS
13	Mechanics	What is the formula for gravitational potential energy?	100%	100%	PASS
14	Optics	What is Snell's law and how does it describe refraction...	100%	100%	PASS
15	Thermodynamics	What is the ideal gas law?	100%	100%	PASS
16	Electromagnetism	Explain what electric potential energy and electric pot...	100%	100%	PASS
17	Mechanics	What is the Doppler effect for sound?	100%	100%	PASS
18	Quantum	What is the Schrödinger equation and what does it descr...	33%	99%	WARN
19	Out Of Scope	What is the best recipe for chocolate cake?	—	—	REFUSED
20	Out Of Scope	Who won the FIFA World Cup in 2022?	—	—	REFUSED

System Architecture

The Physics RAG Chatbot implements a five-stage Retrieval-Augmented Generation pipeline grounded entirely in trusted textbook sources, eliminating LLM hallucination on undergraduate physics topics.

1. Ingestion	PyMuPDF parses 6 PDFs (Feynman Lectures + OpenStax). Text is cleaned and split into 512-character chunks with 64-char overlap, preserving book/chapter/page metadata.
2. Embedding	BAAI/bge-small-en-v1.5 (384-dim, local CPU) converts each chunk to a dense vector. Embedding runs once during ingestion and is stored persistently.
3. Retrieval	MMR (Maximal Marginal Relevance) retrieves the top-10 diverse candidates from ChromaDB using cosine similarity.
4. Reranking	cross-encoder/ms-marco-MiniLM-L-6-v2 scores all 10 candidates, returning the top-3 with a normalised confidence score (sigmoid of raw logits). Queries scoring below 0.35 are refused.
5. Generation	Ollama Llama 3.1 (local) receives the system prompt, retrieved context as [Source N] blocks, and chat history. The Citation Engine maps [Source N] tags in the output back to book/chapter/page metadata.

Corpus

Book	Volumes	Source	Corpus ID
The Feynman Lectures on Physics	Vol 1 (Mechanics) Vol 2 (E&M) Vol 3 (QM)	caltech.edu	feynman
OpenStax University Physics	Vol 1 (Mechanics) Vol 2 (Thermo/Waves) Vol 3 (Optics/Modern)	openstax.org	openstax

Configuration Used

Parameter	Value
LLM	Llama 3.1 (Ollama, local)
Embedding Model	BAAI/bge-small-en-v1.5 (384-dim)
Reranker	cross-encoder/ms-marco-MiniLM-L-6-v2
Vector Store	ChromaDB 1.0.12 (persistent)
Chunk Size	512 characters / 64 overlap
Retrieval Top-K	10 candidates → top-3 after reranking
OOS Threshold	0.35 (reranker confidence)

Embedding Device	CPU
LLM Temperature	0.1